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Review

A Comprehensive Review of Satellite Orbital Placement and Coverage Optimization for Low Earth Orbit Satellite Networks: Challenges and Solutions

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Abstract

Nowadays, internet connectivity suffers from instability and slowness due to optical fiber cable attacks across the seas and oceans. The optimal solution to this problem is using the Low Earth Orbit (LEO) satellite network, which can resolve the problem of internet connectivity and reachability, and it has the power to bring real-time, reliable, low-latency, high-bandwidth, cost-effective internet access to many urban and rural areas in any region of the Earth. However, satellite orbital placement (SOP) and navigation should be carefully designed to reduce signal impairments. The challenges of orbital satellite placement for LEO include constellation development, satellite parameter optimization, bandwidth optimization, consideration of signal impairment, and coverage optimization. This paper presents a comprehensive review of SOP and coverage optimization, examines prevalent issues affecting LEO internet connectivity, evaluates existing solutions, and proposes novel solutions to address these challenges. Furthermore, it recommends a machine learning solution for coverage optimization and SOP that can be used to efficiently enhance internet reliability and reachability for LEO satellite networks. This survey will open the gate for developing an optimal solution for global internet connectivity and reachability.

Keywords: satellite placement; LEO; optimization parameters; coverage optimization

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1. Introduction

Recently, the war in the Red Sea and other locations around the world has affected the main submarine communications cables, resulting in a catastrophic impact on internet connectivity and reachability. Consequently, industrial development and economic growth cannot meet a variety of data communication needs using the current submarine communications cables; hence, a new internet solution is vital. LEO satellite networks offer distinct advantages over traditional terrestrial infrastructures, such as fiber-optic and mobile networks. They provide optimal coverage area and high bandwidth to enhance global internet connectivity, particularly in remote or underserved regions where laying physical infrastructure is challenging. The LEO satellite network is considered an important solution to bring real-time, reliable, low-latency, high-bandwidth, cost-effective internet service in many urban and rural areas on Earth. In recent years, several commercial companies have developed LEO satellite constellations comprising hundreds or even thousands of satellites [1–4]. For instance, SpaceX's Starlink, OneWeb (now part of Eutelsat), and Telesat's Lightspeed network. Some of these constellations have already achieved partial deployment, with Starlink operating over 7000 satellites and OneWeb deploying 648 satellites as of early 2025 [5].

Satellite orbital optimization focuses on selecting the most suitable orbital parameters which are known as Keplerian elements to define the orbits of a satellite constellation and to determine the optimal placement of satellites within those orbits. It can be divided into two categories, which are performance-based optimization and feasibility-based optimization [6]. Performance-based optimization ensures that the orbital configuration is optimized in terms of satellite count, orbital parameters, and coverage patterns to meet the desired performance criteria. Hence, it enhances global communication, improves Earth observation capabilities, and ensures precise navigation services. Feasibility-based optimization involves the practical and operational aspects of the satellite constellation system. It addresses the constraints like budget limitations, technical feasibility, and adherence to international space regulations.

1.1. Satellite Constellation Architecture

In particular, the LEO satellite constellation system has become an important zone for satellite internet providers. It is composed of separated orbits with altitudes of 200 km to 2000 km above the ground. Each orbit has a specific number of satellites, as shown in Figure 1. The satellite networks consist of K orbits, which have the same $360^\circ/(2 \times K)$ angular separation from one another [7,8]. Each orbit has L satellites that only cross each other at the North and South poles. Satellites in the same orbit have an angular separation of $360^\circ/L$. Because the orbits are circular, the radii of satellites and their distances from the same orbital plane always remain constant. According to the second law of motion, the relation between the orbital velocity (V) to the orbital radius (r) can be expressed as

$$V = \sqrt{\frac{G \times M}{r}} \tag{1}$$

where G is the universal constant of gravitation, which equals 6.67259×10^{-11} $m^3kg^{-1}s^{-2}$, and M is the mass of Earth, which equals 5.9736×10^{24} kg.

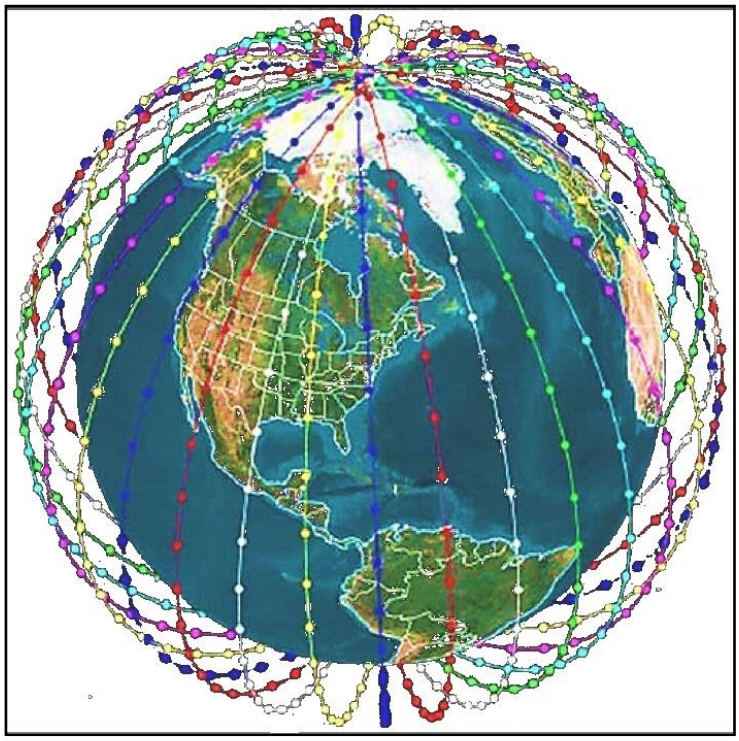


Figure 1. LEO satellite constellation architecture.

The mean orbital velocity for each satellite that is needed to maintain a stable LEO is about 7.8 km/s, which translates to 28,000 km/h. However, this depends on the exact altitude of the satellite's orbit. Therefore, the satellite velocity at a circular orbit with an altitude of 200 km should be maintained at 7.79 km/s and reduced to 6.9 km/s at a circular orbit with an altitude of 2000 km [9]. To provide continuous real-time services, satellites must overlap to cover the whole Earth's surface. The satellites are connected via Inter-Satellite Links (ISLs), which can link satellites in the same or neighboring orbits. We refer to connecting satellites in the same orbit as intra-orbital ISLs. ISLs may also be used to connect satellites with different orbits, which are called inter-orbital ISLs. We presume that intra-orbital ISLs provide more reliability and persistence than inter-orbital ISLs. The horizontal paths of satellites in different orbits vary depending on the location of the satellites. For instance, if the satellites are over the equator, the horizontal distances are longest and shortest when they are over the boundaries of the polar regions. In contrast, the vertical paths of the satellites within the same orbital plane have fixed distances between them throughout the whole orbital connection.

1.2. Operation of LEO Satellite Networks

The modern LEO satellite networks are divided into three parts: LEO space, ground station transceiver, and user devices, as illustrated in Figure 2. The LEO space consists of several orbitals with each containing a fixed number of satellites that communicate using intra-orbital and inter-orbital ISL. Also, the satellites in each orbit can communicate with the nearest ground stations that are distributed on Earth. Moreover, the ground segment consists of several ground stations (GSs) that are distributed on the Earth to monitor, manage, and control the platforms and the signals sent by the satellites. In addition, the GSs are connected to the partnering mobile network operators to provide internet services for the subscriber users. The mobile network operators might utilize LTE or 5/6G communication technology, which depends on available infrastructure. Finally, the user segment consists of user devices, wireless broadband routers, and network applications that can serve smart mobile phones, iPads, tablets, PCs, servers, vehicles, IoT, and sensor devices, etc. It is interesting to know that the GSs typically do not interact directly with the user segment, but only with the space segment.

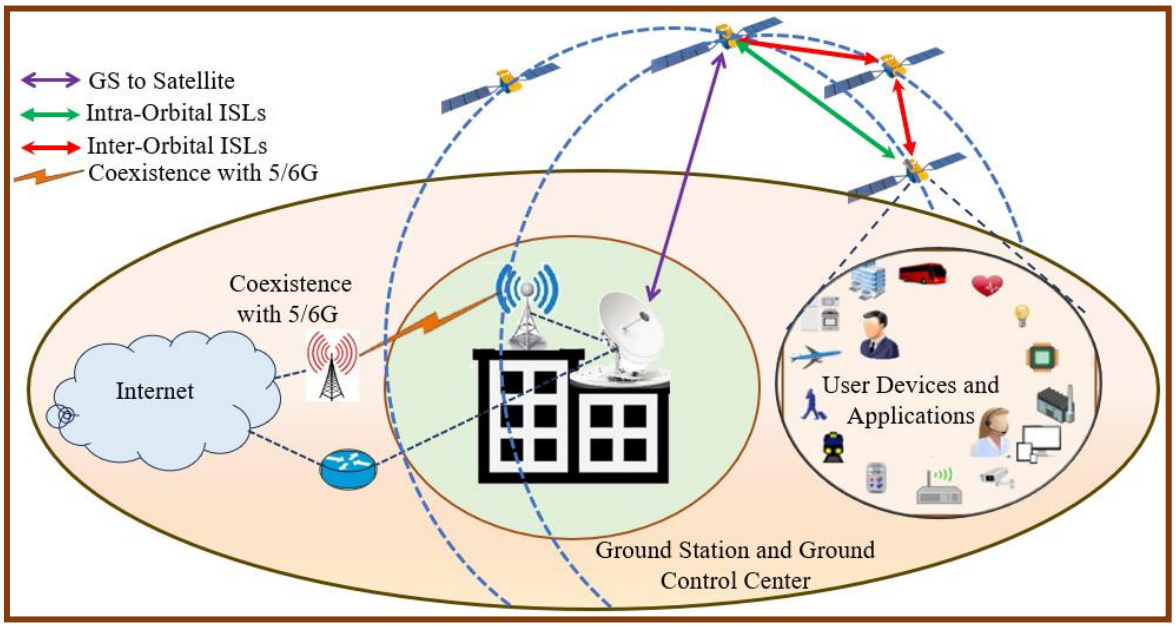


Figure 2. Coexistence of LEO satellite networks and terrestrial networks.

The communication process in the modern LEO satellite networks is described as follows:

- Direct Connection: The user device connects directly to the satellites that cover the user device's area.
- Satellite Relay: The satellites are equipped with an onboard LTE/5/6G modem (eNodeB) which acts like a cell tower in space that can receive from and transmit to user devices.
- Inter-Satellite Links (ISLs): The satellite uses laser ISLs to transmit the signal across the satellite network toward the satellite that is positioned over a GS.
- Ground Station Link: The signal is then downlinked to a GS connected to terrestrial networks that can provide an internet backbone.

The trend of ubiquitous connectivity is the coexistence between LEO and 5G mobile communication networks, which has become an increasingly pressing issue for the satellite networks industry. The coexistence between 5/6G and LEO will facilitate internet access and voice services in mountains, sea, sky, islands, and remote rural areas. However, several other obstacles must be overcome to guarantee that these technologies function properly without significantly degrading co-channel communications; for example, the interference impact from 5G user equipment and base station on LEO satellite signal, the continuous adjustment of communication links, and the unique routing challenges. As a result, the network topology frequently changed due to satellite movements, which did not guarantee the reliability of the established connections in the network. Moreover, communication between LEO and ground stations experiences potential signal blockage due to tall buildings, and a limited coverage area [10–12]. Thus, optimizing SOP is required to create the optimal and reachable path between the source and the destination and to maintain reliable internet connectivity.

1.3. Problem Statement and Research Motivation

The presence of space debris and defunct satellites poses a constant threat to active satellites, as even small fragments can cause significant damage due to their high velocities. Furthermore, the absence of advanced technologies to retrieve or deorbit non-functional satellites and space debris poses a crucial challenge for space internet provider companies to maintain safe orbital environments. The optimal SOP ensures effective service delivery, such as broadcasting, navigation, and internet connectivity. However, the satellite channel in the lower orbital placement suffers from rapid signal decay, which is caused by atmospheric drag (gases), obstacles, and signal noise. Higher orbital placement causes higher latency (from 0.67 ms at 200 Km to 6.7 ms at 2000 Km) and signal decay and low signal strength. This means that the highest SOP provides unreliable internet connectivity. The main motivation for this research is to review the optimization of SOP, which presents significant challenges. Factors such as orbital altitude, inclination, and spacing directly impact coverage, latency, and network resilience.

1.4. Research Contribution

The following contributions are reported in this research:

- It reviews the optimization methods of SOP and coverage for LEO networks. It highlights the main parameters that have been considered in increasing internet connectivity, coverage area, and reliability.
- It analyses the LEO parameters that can mitigate the effect of signal impairments and space debris, and it enhances the swath width for the LEO satellite network.
- It suggests two solutions for SOP and coverage optimization for LEO networks based on effective machine learning algorithms.

The rest of this paper is arranged as follows: Section 2 presents related works on SOP, coverage optimization, and signal impairments. The analysis of SOP and coverage optimization is explained in Section 3. Also, Section 4 describes the proposed solutions based on machine learning algorithms, and Section 5 explains the limitations and future work. Finally, Section 6 presents the conclusion and future work.

2. Related Works

In recent years, many global internet service providers have deployed thousands of satellite constellations in LEO due to lower cost, smaller size, feasibility, and global investment. Therefore, some companies support many researchers who have looked at various issues in SOP, which include efficient LEO coverage area, signal strength quality and impairments, handover, latency, and the number of satellites required to maintain internet reliability and reachability. Commonly, there are several trends of optimizing solutions for SOP that have been proposed in the literature as follows.

2.1. Related Work on Satellite Orbital Placement

The optimization of orbital placement in LEO satellite networks is essential for enhancing coverage, reducing latency, and ensuring network resilience. Several studies have been proposed in the literature to address the inherent challenges in designing efficient LEO constellations. Z. Shang [13] utilized deep reinforcement learning in optimizing satellite constellations, which significantly improved the communication speed and coverage efficiency in key regions such as underdeveloped areas and maritime routes, where deploying ground stations is challenging. The authors adjusted satellite orbital layers and densities to optimize the utilization of existing resources without increasing the total number of satellites. Also, F.S. Prol et al. [14] reviewed several requirements to build an LEO positioning, navigation, and timing (PNT) system (LEO-PNT) that includes the transmitting signal between the space segment, ground segment, and user segment. However, the authors have not recommended any desirable selections in every single aspect of the LEO-PNT system because of the lack of simulations in the current literature. G. Haibo et al. [15] used statistical analysis in terms of the number of orbitals, the number of satellites in each orbital, and the inclination angle to optimize LEO constellation for the Global Navigation Satellite System (LeGNSS). The authors in [15] demonstrated that combining multiple LEO satellite constellations with varying orbital inclinations results in a more uniform distribution of visible satellites across different latitudes. Moreover, Kaan et al. [16] presented an overview of the LEO optimization methods for the three parts in the LEO system network, including communication, navigation, and/or sensing applications. The authors in [16] compared various optimization methods in terms of their complexity, convergence characteristics, and feasibility. They presented the impact of orbital altitudes on latency and the significance of orbital plane positioning in determining coverage areas. Also, L. Jing et al. [3] designed two hybrid LEO constellation systems based on the NSGA-III optimization algorithm to enhance the BeiDou Satellite Navigation System (BDS). The proposed constellations in [3] reduced the number of satellites involved by more than 100. Furthermore, G.Junqi et al. [17] proposed a fast satellite selection algorithm for positioning in the LEO constellation, which used a geometric method to select an optimal subset of satellites that meet the different numbers of observation satellites and is close to the optimal geometric configuration. Also, Kaan et al. [1] studied the design of the LEO-PNT constellation optimization, and the optimization approaches of the state-of-the-art for LEO satellite constellation, including the effective metrics and their performance that must be considered for any LEO-PNT system design. For instance, studies have proposed methodologies that consider parameters such as orbital inclination, number of satellites, and orbital planes to enhance precise point

positioning solutions. In addition, I.F.Ghoniem et al. [18] proposed a GNSS-LEO optimization for Egypt and the Middle East region. The authors used two GNSS constellations data which are GPS with 29 satellites and GLONASS with 24 satellites. They studied more than a hundred orbit cases for Middle East region, and they found that the most optimal parameters for satellite placement are altitude = 1500 km, inclination = 0°, initial value of mean anomaly = 330°, and the other parameters are set to 0 values. The most related work is proposed by Yuta et al. [19], who suggested a metaheuristic optimization method for constellation determination using a mathematical formulation, focusing on orbit requirements for interferometric applications. W. Xue et al. [20] studied the design of LEO navigation constellations while considering the task requirements of different stages of constellation deployment. They suggested solutions for performance degradation issues. Table 1 summarizes the tools and methods used, problems addressed, and outcomes achieved in the previous literature studies [1,3,13–20].

Table 1. Summary of SOP: tools and methods, problems tackled and results obtained.

Research Work (Year)	Tools/Methods	Problem Solved	Effects Achieved	Limitation (Analytical)
Z. Shang [13] (2024)	Deep reinforcement learning	Optimization of satellite constellation for communication coverage in remote zones	Increased link throughput (~500 Mbps) and improved coverage in challenging regions	Region-specific optimization constrains scalability; the model may require substantial retraining when applied to different orbital inclinations, latitudes, or ground infrastructure conditions.
F.S. Prol et al. [14] (2023)	NSGA-III multi-objective GA	Reduced satellite count for BeiDou enhancement while maintaining coverage	Achieved hybrid constellations (177 and 186 sats) with improved GDOP and positioning	Limited simulation datasets risk overfitting to idealized scenarios, making performance uncertain under variable environmental or operational conditions.
G. Haibo et al. [15] (2025)	Literature review of LEO-PNT systems	Identification of system design and signal requirements for LEO-PNT	Compiled a comprehensive design framework	The absence of environmental factor modeling (e.g., atmospheric interference, orbital debris) limits applicability for dynamic and unpredictable real-world orbital conditions.
Kaan et al. [16] (2022)	Orbital parameter analysis	Promoted uniform satellite distribution for GNSS constellations	Improved constellation uniformity across latitudes	Focuses on parameter-driven uniformity but does not integrate emerging AI/heuristic optimization methods, potentially missing more adaptive deployment strategies.
L. Jing et al. [3] (2021)	Geometric satellite selection algorithm	Fast selection of optimal satellites for constellation deployment	Higher computational efficiency in selection	Prioritizes computational speed, which may sacrifice redundancy and robustness, especially in fault-tolerant or mission-critical applications.
G. Junqi et al. [17] (2023)	Comparative optimization methods	Evaluated tools and metrics across communication, navigation, and sensing domains	A broader understanding of method suitability	Lack of validation under real-world signal degradation scenarios weakens confidence in long-term operational reliability.
Kaan et al. [1] (2023)	Regional orbital optimization (100+ orbital cases)	GNSS-LEO optimization for the Middle East	Identified an optimal regional orbit set (altitude 1500 km, inclination 0°)	Constrained to Middle Eastern regional parameters, performance in other regions may vary significantly due to different orbital dynamics and coverage demands.
I.F. Ghoniem et al. [18] (2020)	Metaheuristic optimization	Constellation design for interferometric applications	Found effective solutions for interferometric performance	Geographic specialization restricts broader adoption; the algorithm may require significant adaptation for diverse orbital regimes.

Table 1. *Cont.*

Research Work (Year)	Tools/Methods	Problem Solved	Effects Achieved	Limitation (Analytical)
Yuta et al. [19] (2010)	Deployment-phase aware design	Optimization based on constellation deployment stages	Tailored designs to mission requirements across deployment stages	Metaheuristic complexity imposes high computational demands, which may hinder real-time decision-making in dynamic mission environments.
W. Xue et al. [20] (2024)	Deep reinforcement learning	Optimization of satellite constellation for communication coverage in remote zones	Increased link throughput (~500 Mbps) and improved coverage in challenging regions	Limited testing under performance degradation scenarios leaves uncertainty about resilience in adverse signal and environmental conditions.

2.2. Related Work on Optimization of LEO Coverage Area

In this literature, several recent studies have used AI-driven optimization, beamforming techniques, constellation design, and resource allocation strategies to optimize the coverage areas in LEO satellite networks. The most related work is proposed by S. Cakaj [21,22], who developed a mathematical model for LEO coverage belt and found that orbit attitudes between 5633 and 8177 km and elevation angles between 2 and 10° achieve the wideness coverage area. Also, Z. Titus et al. [23] proposed a mathematical model for optimization of multi-altitude LEO satellite networks to achieve efficient coverage. The authors used metrics such as coverage probability, signal strength, interference levels, capacity, and quality of service to assess the performance of the coverage area based on the Cox point process model and the optimized satellite deployment. The recent related work is proposed by Silvirianti et al. [24], who suggested a quantum adaptive learning (QAL) based on the advantage of quantum computing and adaptive learning as a potential solution for coverage optimization of stochastic geometry-based LEO satellite networks. Also, J. Shin et al. [25] proposed discontinuous regional coverage for LEO based on analytical constellations design, which determines the best inclination for a given constellation that maximizes the coverage for multi-ground regions. The authors in [25] developed an optimal inclination search algorithm which considers both the region location and the satellite coverage range. Also, I.Lluch et al. [26] developed an optimal approach for satellite-to-satellite coverage and its analytical validation. The authors have shown that increasing the inter-satellite-link maximum range above 6000 km does not lead to further coverage benefits at LEO altitude. Miyeon et al. [27] proposed mathematical analyses in terms of cluster area, LOS intensity, and SIR threshold for efficient coverage probability. The mathematical model proposed in [27] is aiding in reliable satellite cluster network design, where satellites in the cluster area collaborate to serve users in mega-constellations. Peng Zong et al. [28] used a genetic algorithm to propose an optimized coverage of a constellation satellite in one revisit and the regional coverage at a defined latitude. The authors in [28] have shown that the Earth's pole can be covered if the inclination angle of the satellites is more than 90 degrees. The optimal altitude can be used to provide a revisit time between 90 and 130 min, which means that satellites can cover the entire Earth in one revisit. Hassan et al. [29] proposed reconfigurable intelligent surfaces (RISs) within 6G sub-THz networks to maximize LEO satellite coverage, which is used to ultimately maximize end-to-end data rate through optimizing network performance that includes satellite-RUE association, data packet routing in satellite constellations, RIS phase shift, and GBS transmit power. Also, Zhao et al. [30] proposed an autonomous self-healing framework that captured the multi-objective of maximizing coverage performance, minimizing the total control effort for satellite constellation adaptation, and improving the resilience of the satellite constellation coverage for adversarial and non-adversarial attacks. Table 2 summarizes the tools

and methods used, the problems addressed, and the outcomes achieved in the previous literature studies on the optimization of LEO coverage areas [21–30].

Table 2. Summary of optimization of LEO coverage area: tools and methods, problems solved, and results obtained.

Research Work (Year)	Tools/Methods	Problem Solved	Effects Achieved	Limitation (Analytical)
S. Cakaj [21,22] (2016, 2014)	Analytical modeling of LEO belt coverage	Identification of optimal altitude (5633–8177 km) and elevation angles (2–10°)	Defined parameter ranges that maximize the ground coverage area	Static analytical assumptions overlook dynamic environmental and operational constraints, which can lead to deviations from predicted coverage performance in real-world deployments.
Z. Titus et al. [23] (2023)	Cox point process + GA-based optimization	Multi-altitude constellation design optimizing coverage probability, capacity	Derived performance metrics (coverage probability, interference) and optimized altitude allocations	High computational complexity may hinder real-time adaptability and scalability, especially for large-scale constellations or rapidly changing traffic patterns.
Silvirianti et al. [24] (2025)	Quantum adaptive learning (QAL) + stochastic geometry	Coverage optimization in stochastic LEO network configurations	Promising adaptive coverage enhancement leveraging quantum algorithms	Reliance on emerging quantum computing technologies limits near-term applicability due to hardware immaturity and high integration costs.
J. Shin et al. [25] (2021)	Analytical constellation design + inclination search	Regional, discontinuous LEO coverage specific to target regions	Improved regional coverage through optimal inclination selection	Regional focus limits the applicability to global systems; the approach lacks mechanisms for rapid reconfiguration to respond to shifting coverage demands.
I. Lluch et al. [26] (2014)	Satellite-to-satellite link modeling	Determination of optimal inter-satellite link range at LEO altitudes	Showed diminishing returns beyond ~6000 km link ranges	Does not incorporate advances in laser or phased-array inter-satellite communications, potentially underestimating achievable link distances in future systems.
Miyeon et al. [27] (2025)	SIR/LOS threshold modeling + mathematical analysis	Enhancing coverage probability in clustered satellite networks	Identified optimal SIR/LOS thresholds to boost effective coverage	Requires extensive empirical data for calibration; may struggle to adapt to fast-changing link quality in dynamic orbital environments.
Peng Zong et al. [28] (2019)	Genetic Algorithm (GA)	Single-revisit regional coverage optimization	Demonstrated pole coverage using inclinations > 90° with minimal satellites	Optimizes revisit frequency but does not fully address trade-offs between coverage persistence, latency, and system resource allocation.
Hassan et al. [29] (2024)	RIS-enabled 6G + sub-THz networks	Integration of LEO with terrestrial 6G to maximize joint coverage	Theoretical gains in coverage enhancement using RIS technology	Dependent on RIS technology readiness and its seamless integration into heterogeneous satellite–terrestrial network architectures.
Zhao et al. [30] (2022)	Autonomous self-healing framework	Maximizing coverage while minimizing control effort and system resilience	Showed resilient coverage adaptation under perturbations	Limited validation across diverse operational scenarios; scalability and interoperability with existing network control frameworks remain uncertain.

2.3. Related Work on LEO Orbital Signal Impairments

Several recent studies have addressed various challenges of signal impairments in LEO satellite networks, which include jamming, interference, atmospheric effects, and hardware limitations, along with their proposed solutions and noted limitations. Christina et al. [31]. surveyed receiver designs for LEO satellite signals, discussing challenges like Doppler shifts and signal attenuation, and exploring augmentation methods such as Satellite Timing and Location (STL) services. Also, Qian Ning et al. [32] proposed the system model of shadow fading and rain attenuation in the satellite downlink channel for LEO. The authors analyzed the application of Non-Orthogonal Multiple Access (NOMA) in LEO satellite

communications under rain attenuation and fading, demonstrating improved ergodic capacity. The most related work was proposed by JIA Min et al. [33], who investigated inter-satellite link interference in large-scale LEO constellations, analyzing attenuation characteristics and time–frequency distributions. The authors in [33] used simulation to prove that there is a noticeable interference in the higher frequency links among large-scale LEO satellite constellation systems. Furthermore, Jiawei Liu et al. [34] proposed an approach for in-orbit calibration of the phase-center offsets (PCOs) and code hardware delays of the LEO downlink navigation signal, which enhanced signal accuracy. The finding of [34] was that increasing the number of tracking stations and processing periods can improve the formal precision of PCOs and hardware delay. Also, A.K Dwivedi et al. [35] studied two interference systems at the ground station to mitigate interference, which are successive interference cancelation (SIC) and captured model (CM)-based decoding schemes. The average outage probability for the CM-based and SIC are derivative analytically under an extreme signal-to-noise ratio (SNR), which are utilized to optimize the system parameters for achieving a target outage probability. Moreover, R.Miteva et al. [36] presented two space weather phenomena, which are geomagnetic storms and solar flares, that have a high impact on satellite operations. They discussed how sequences of geomagnetic disturbances, even if individually weak, can cumulatively lead to significant atmospheric drag, potentially resulting in satellite service disruptions or losses. Also, Radojkovic et al. [37] presented the impact of gamma-shadowed Ricean fading on the secrecy capacity of LEO satellite-to-ground communications and analyzed the secrecy performance of an LEO satellite and ground user (U) downlink in the presence of an eavesdropper, over Gamma-shadowed Ricean fading channels. In addition, Bassel F. Beidas [38] proposed an effective I/Q imbalance introduced by analog frequency-conversion circuits in LEO satellite systems which used a digital compensation algorithm with immunity to frequency offset. Ji Ma et al. [39] proposed a resilience measure for the LEO satellite networks, which utilized uncertainty theory to define belief in instantaneous availability. The authors in [39] developed an uncertain satellite network evolution model that considered various impairments to describe the operating pattern of dynamic LEO and to estimate resilience. Zhenghao Zhang [40] proposed a packet acquisition method using chirp signals to improve detection in weak signal environments and reduce peak-to-average power ratio. Table 3 summarizes the tools and methods used, problems solved, and outcomes achieved in the previous literature studies [31–40].

Table 3. Summary of related works LEO orbital signal impairments: tools, issues, and results.

Research Work (Year)	Tools/Methods	Problem Solved	Effects Achieved	Limitation (Analytical)
Christina et al. [31] (2023)	Receiver design survey, augmentation methods (e.g., STL services)	Doppler shift, signal attenuation in LEO downlink	Identified common challenges and potential STL-based mitigation strategies	STL services inherently offer lower accuracy than GNSS, and their limited geographic availability restricts global applicability of the proposed augmentation solutions.
Qian Ning et al. [32] (2022)	Hybrid channel modeling (rain + shadow fading) + NOMA techniques	Rain and shadow fading impact on LEO downlink capacity	NOMA improved ergodic capacity under fading; lower altitude and power usage increased efficiency	Gains are highly dependent on accurate channel state information; real-time estimation may be challenging, and implementation complexity could hinder large-scale adoption.
JIA Min et al. [33] (2022)	Simulation of inter-satellite interference in LEO	Signal interference across large LEO constellations	Demonstrated significant interference in high-frequency inter-satellite links	Exclusively simulation-based, which may fail to capture hardware imperfections, atmospheric effects, and operational variability observed in real deployments.

Table 3. *Cont.*

Research Work (Year)	Tools/Methods	Problem Solved	Effects Achieved	Limitation (Analytical)
Jiawei Liu et al. [34] (2024)	In-orbit calibration of phase-center offsets (PCOs) and hardware delays	Downlink signal accuracy degraded by hardware/phase-center offsets	Calibration approach enhanced signal precision via ground tracking	Requires dense ground station networks and continuous observational data; precision may degrade in regions with sparse tracking infrastructure.
A.K. Dwivedi et al. [35] (2023)	Analytical outage probability modeling + SIC/CM decoding schemes	Ground station decoding under interference and high SNR	Derived formulas for outage probability, enabling optimized system design	Assumes perfect synchronization, which is rarely achievable in real-world scenarios; asynchronous transmissions could reduce decoding reliability.
R. Miteva et al. [36] (2023)	Space weather survey (geomagnetic storms, solar events)	Atmospheric drag and operational resilience during space weather events	Highlighted cumulative weather effects potentially causing major service disruption	Focuses primarily on cumulative long-term impacts; immediate, short-term mitigation strategies for active satellite operations are not fully addressed.
Radojkovic et al. [37] (2025)	Secrecy capacity modeling under gamma-shadowed Ricean fading	Secure LEO downlink under complex fading and eavesdropping threats	Quantified secrecy capacity reductions under specialized fading channel	Assumes a narrow set of fading models, potentially limiting applicability to environments with mixed or non-stationary channel conditions.
Bassel F. Beidas [38] (2022)	Digital frequency offset compensation, I/Q-imbalance correction	Frequency offset and I/Q errors in LEO analog frequency converters	Improved signal fidelity through digital algorithms	Algorithm effectiveness may degrade under hardware variability and temperature drift; integration requires complex calibration procedures.
Ji Ma et al. [39] (2022)	Uncertainty theory-based resilience modeling	Network resilience against multiple impairments	Proposed belief-based availability measure and evolution model	Requires extensive, high-quality datasets to accurately calibrate uncertainty models; data scarcity in certain operational contexts could limit reliability.
Zhenghao Zhang [40] (2024)	Chirp-based packet detection under weak signals	Packet detection in low-SNR LEO environments	More reliable packet acquisition with reduced peak-to-average power	Performance sensitivity to varying noise profiles and Doppler rates may reduce consistency; compatibility with existing receiver architectures needs assessment.

3. Analysis of Satellite Orbital Placement and Coverage Optimization for LEO

The optimization of SOP in LEO is fundamental for enhancing coverage area, reducing latency, and ensuring the efficiency of satellite networks. This section investigates the key parameters and methodologies involved in optimizing LEO SOP.

3.1. Orbital Parameters and Their Impact

Several orbital parameters have been investigated in the literature studies, including altitude, latitude, inclination angle, elevation angle, swath width, packet latency, power of transmission, space debris, and SINR.

3.1.1. Satellite Coverage Area

The ground coverage area of a satellite can be assumed as a circle on the Earth with the elevation angle (ε) between 0 and 90° as can be shown in Figure 3. The radius of the coverage area (D) for a satellite A, which has been assumed to have a line-of-sight (LoS) constraint defined by ε, can be calculated using the following equation [16,28]:

$$D_{CA}(\varepsilon) = R_{earth}\left[\sqrt{\left(\frac{H + R_{earth}}{R_{earth}}\right)^2 - \cos^2(\varepsilon)} - \sin(\varepsilon)\right] \quad (2)$$

where R_{earth} = 6378 Km, and H is the orbital altitude. The coverage angle (Θ) of a satellite is an angle that specifies the area on the ground surface that allows the satellite to communicate with the devices on the ground surface at any given moment. The actual surface area that is covered by satellite A (the area of a spherical cap) can be calculated as follows:

$$C(A) = 2\pi R_{earth}^2(1 - \cos(\theta)) = 2\pi R_{earth}^2\left(1 - \cos\left(\frac{D_{CA}(\varepsilon)}{R_{earth}}\right)\right) \quad (3)$$

Since the radius $D_{CA}(\varepsilon)$ is smaller compared to the R_{earth}, Equation (3) can be approximated to

$$C(A) = 2\pi(D_{CA}(\varepsilon))^2 \quad (4)$$

Equation (4) facilitates optimizing communication visibility and internet reachability. However, satellite coverage can be influenced by factors such as atmospheric attenuation, obstacles blocking transmission, and signal impairment. It is interesting to know that higher orbital altitudes (H) will increase the coverage area but may reduce signal strength and increase latency. Also, higher ε will reduce the coverage area but improve signal quality by reducing atmospheric interference.

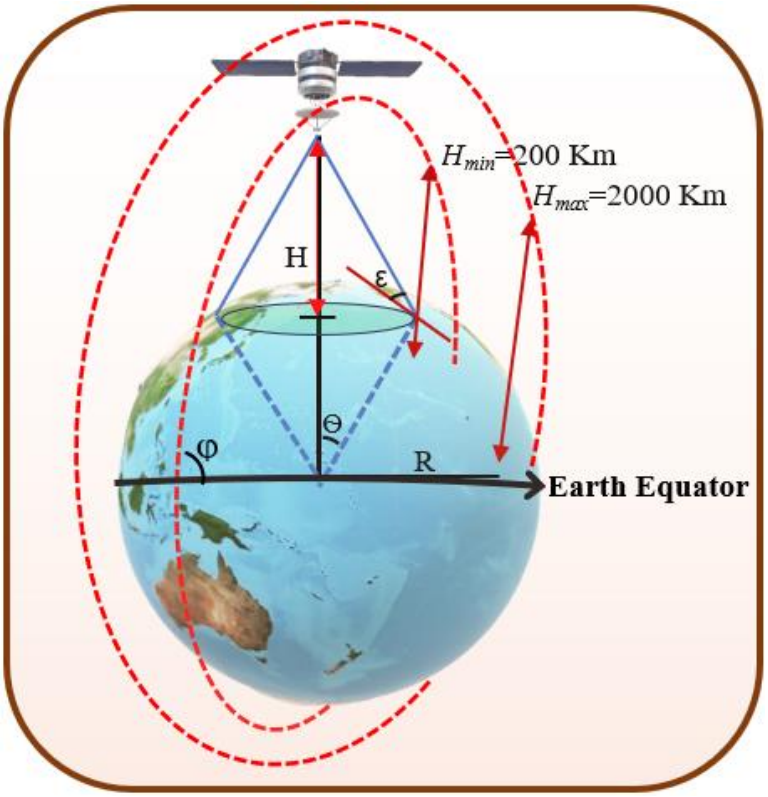


Figure 3. LEO satellite coverage area.

3.1.2. Satellite Swath Width

The LEO placement can be used in imaging, sensing, and observation planning. For this purpose, the satellites are equipped with sensors or high-resolution cameras.

The inclination angle (φ) determines the latitude limits of the orbit which specifies the relevant area for overall orbital coverage. Moreover, the φ with 60° to 120° is considered a polar orbit, which enables the satellite to be able to view a larger fraction of the Earth. As shown in Figure 4, the swath width refers to the strip of the Earth's surface from which geographic data is collected by a satellite. It is the width of the light strip on the surface below as the flashlight moves. The nadir line is a straight vertical line downwards and

perpendicular to the Earth's surface. Also, the angle θ_H in radian is the angle measured at the satellite, which is placed between the nadir line and the line that has been drawn to the left edge of the camera/sensor field of view (FOV). The equation of swath width (SW) in the curved Earth model can be derived as follows [41]:

$$SW = 2 \times R_{earth} \times \left(\sin^{-1} \left(\frac{(R_{earth} + H) \times \sin(\theta_H)}{R_{earth}} \right) - \theta_H \right) \quad (5)$$

While the coverage area Equation (2) is used to measure communication visibility, the swath width Equation (5) is used to measure imaging and for observation planning.

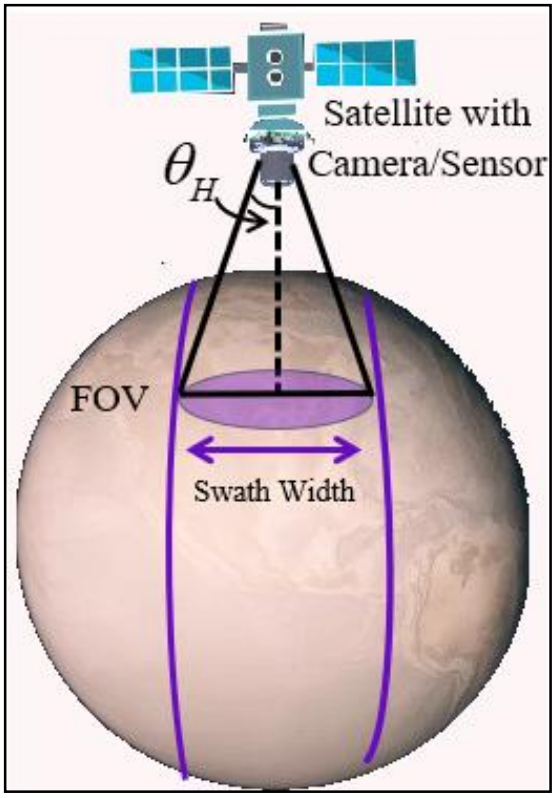


Figure 4. Swath width for satellite camera/sensor.

3.1.3. LEO Space Debris

Space debris refers to any unused objects that are orbiting Earth and were created from defunct satellites, spent rocket stages, fragments from explosions or collisions, and even paint flecks. The danger of space debris lies in high velocities of objects in LEO (up to 28,000 km/h), which means that even small pieces of debris can cause significant damage to operational satellites or spacecraft, which leads to loss of function or even catastrophic breakups. This creates a cascading effect known as Kessler Syndrome, which concludes that collisions generate more debris and increase the likelihood of further collisions. As a result, space debris presents a considerable risk to both present and forthcoming space operations, which potentially affects critical services such as telecommunications, global positioning systems, Earth surveillance, and manned space exploration. LEO is particularly crowded with two types of debris: tracked objects (>10 cm) and untracked objects (<10 cm). As of early 2025, the number of tracked objects in LEO is approximately 22,500, which includes 11,700 operational and nonoperational payloads, 950 rocket bodies, 7750 debris fragments, and 2100 unknown objects. Moreover, the number of untracked objects is divided into three categories: 1–10 cm, which is approximately 600,000 pieces; 1 mm–1 cm, about 10 million pieces; and less than 1 mm, more than a trillion pieces [42–46]. As can

be seen in Figure 5 and Table 4, the most congested debris area in LEO space is between 500 and 1000 km with both tracked and untracked objects. The debris area that is located between 800 and 1000 km is called the congested LEO belt. In contrast, the two LEO space areas (200–500 km and 1500–2000 km) have lower debris density. However, the first area has frequent orbital decay due to atmospheric drag. The second area has long delays (10 times longer delays) compared to the first area.

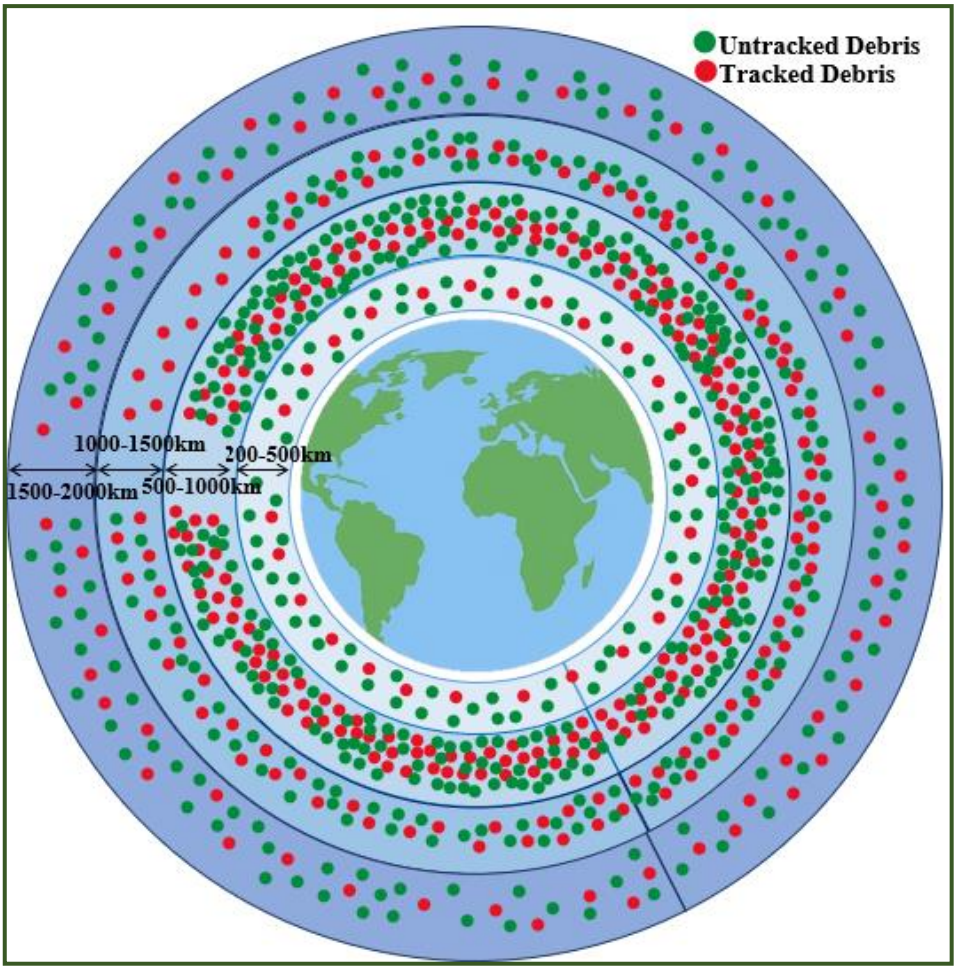


Figure 5. LEO space debris.

Table 4. Tracked and Untracked Debris in LEO.

Altitude Range (km)	Tracked Debris (>10 cm)	Estimated Untracked Debris (1 mm–10 cm)	Description
200–500	~1200	~60,000	Lower density; frequent orbital decay due to atmospheric drag. Debris typically re-enters within years.
500–800	~4500	~150,000	Moderate density; long-lived orbits. Popular for remote sensing missions.
800–1000	~6800	~200,000	Densest zone, called “congested LEO belt”. Site of several breakup events (e.g., Iridium–Cosmos collision).
1000–1500	~3000	~80,000	Includes old satellites and upper stages. Low drag means that the debris remains for centuries.
1500–2000	~1000	~30,000	Sparse compared to lower LEO, but debris is persistent due to low atmospheric influence.

3.1.4. LEO Signal Impairments

Signal impairment is the degradation of a satellite communication signal due to signal distortion, noise, interference, and attenuation. Managing and reducing signal impairment is a primary challenge due to a crowded satellite environment and shared limited frequency bands in the LEO communication system. In addition, if signal impairment is too strong, a satellite ground station or user terminal cannot correctly interpret the data from the satellite, leading to communication failure. Furthermore, signal impairments can come from various space and ground sources:

- Intra-System Interference: Beams from adjacent satellites can overlap, which causes interference for users on the ground.
- Inter-System Interference: Interference from satellites in different LEO constellations or from satellites in higher orbits (MEO/GEO) that use adjacent frequency bands.
- Terrestrial Interference: Signals from ground-based systems such as 5G towers, Wi-Fi networks, and microwave signals are a major source of interference.
- Space Weather: Solar flares and geomagnetic storms cause signal attenuation or noise.
- Intentional Interference (Jamming): Malicious attempts to transmit powerful signals at a satellite to disrupt its communication link.
- Multipath and Doppler Effects: Caused by fast satellite movement and terrain-based signal reflection.

When a satellite signal is affected by any of the aforementioned signal impairments, it leads to several negative consequences for the user and the network. For example, it reduced network data rate (throughput), increased connection bit error rate (BER) and latency, and dropped connections between the satellite and receiver. The primary metric used to measure the quality and clarity of a communication signal in the presence of interference and noise is the **Signal-to-Interference-plus-Noise Ratio (SINR)**, which can be formulated as [47–49]

$$SINR(d) = \frac{Pr(d)}{I_{\mathrm{n}} + N} = \frac{Pr(d_0) + 10\beta \log(\frac{d}{d_0}) + X\sigma}{\sum_{i=1}^{n} I_i = \sum_{i=1}^{n} (Pr_i(d) \times CI_i) + N} \quad (6)$$

where d is the distance between the satellite and the receiver, which could be another satellite or ground station or user device, d_0 is the reference distance, $Pr(d)$ is the received signal, I_{n} presents the total multipath interference, N is the total noises that affect the received signal, and β is the path loss exponent which reflects the signal decay over distance in a specific propagation environment. Xσ is a Gaussian distributed random variable in (dB). ***CI*** is the overlapping factor between 0 and 1, and it depends on the spectral properties (channel width, inter-channel spectral distance, and spectral mask) and the separation from the interferer channel i.

4. Discussion of Satellite Orbital Placement Proposed Solutions

The suggested satellite placement solution addresses the aforementioned problems and selects the optimal effective coverage parameters. In order to overcome space debris, signal impairments, and frequent orbital decay, the optimal placement solution should use an AI algorithm that can manage different variables and consider all effective challenges.

4.1. Satellite Orbital Placement Proposed Solution

This research suggests using machine learning (ML) algorithms to estimate the optimal SOP based on the aforementioned analysis parameters. Among ML algorithms, distributed reinforcement learning (DRL) can accept multifunction inputs and provide optimal target

outputs. This research defines optimal SOP as the state–action outcome that maximizes a weighted, multi-metric reward. The input state includes SINR, latency, altitude, debris density (tracked and untracked), and collision risk. Thus, optimal placement corresponds to policy decisions that deliver the highest long-term cumulative reward. The reward function R(s, a) is computed from critical parameters including SINR, end-to-end delay, optimal altitude, number of tracked and untracked debris counts, coverage gain, and collision risk. The formulation of the reward function can be expressed as follows:

$$R(s,a) = w_1 SINR - w_2 Delay + w_3 CovGain - w_4(1/AltitudeDiff) - w_5 CollisionRisk \quad (7)$$

where the weight w_i is tunable via simulation and domain knowledge. The flowchart diagram of the suggested SOP model based on DRL is illustrated in Figure 6 which performs the following steps:

- Step 1: Define state and action.
  States include current altitude, SINR, latency, debris encounter rates, and estimated collision probability.
  Actions consist of discrete or continuous orbital actions such as altitude adjustments, inclination shifts, and short collision avoidance.
- Step 2: Initialize the matrix Q of DRL.
  The Q-matrix $Q(s,a)$ must be constructed and initialized to 0. The columns represent all possible actions, while the rows represent the states for each action.
- Step 3: Select and perform action.
  For each episode, an agent takes an action a (e.g., adjust altitude or perform collision avoidance), transitions to the next state s', and receives a reward $R(s,a)$.
- Step 4: Reward estimation.
  The reward $R(s,a)$ in Equation (7) is computed.
- Step 5: Q-value update and evaluation.
  The maximum Q-value for the next state is estimated as

$$Q_N(s,a) \leftarrow (1-\alpha)Q_C(s,a) + \alpha\left\{R(s,a) + \gamma Max_{x\in S}\widetilde{Q}_x(s,a)\right\} \quad (8)$$

where α represents the learning rate, γ is the discount factor, Q_N is the new value of Q, Q_C is the current Q-value, and $Max_{x\in S}\widetilde{Q}_x$ is the maximum predicted reward, given new states and possible actions.

- Step 6: Integrated collision avoidance.
  Collision avoidance is embedded within the action space and reward function. Consequently, an AI system with an advanced detection system should be applied for each satellite to perform real-time orbital adjustments.

It is interesting to note that the optimization calculation is estimated at the ground control center at GS, not in the satellite. This means the satellite receives commands from the GS control center and adjusts its position accordingly. Furthermore, most of the satellite equipment is equipped with a solar system that keeps the batteries charged. Hence, the computing power at the satellite is not a critical issue. Furthermore, the convergence time in the proposed system refers to how long the DRL/optimization model takes to learn the optimal policy, not how long satellites take to execute it. This convergence occurs entirely at the ground control center, where high-performance computing (e.g., GPU clusters) is available.

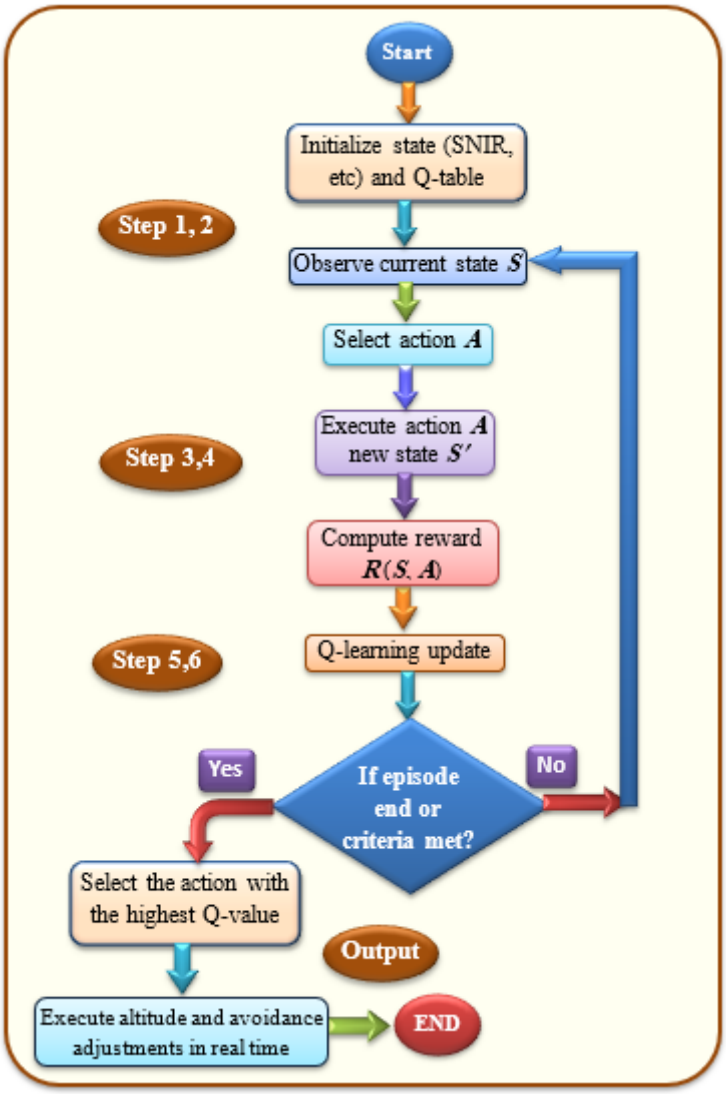


Figure 6. Flowchart diagram for suggested DRL SOP solution.

The final output of the DRL algorithm is a learned Q-table and deployment policy for actions:

- A learned Q-table that maps each state (including current altitude, debris count, SINR, etc.) to action values.
- A policy that chooses the action with the highest Q-value in each state. This translates to recommended orbital placement and operation commands for each satellite agent.
- Select optimal action via the learned policy.
- Execute altitude and avoidance adjustments in real time.

4.2. Coverage Optimization Proposed Solution

The dynamic values of inclination angle, elevation angle, and swath width for each satellite in the same orbit should be utilized to suggest an optimization function. Furthermore, the estimation of SOP is recommended to be involved in the coverage area optimization because it correlates with inclination angle, elevation angle, and swath width calculation. The suggested algorithm follows the following steps:

- Step 1: Obtain the value of SOP.
  The latitude (H) should be estimated from the previous algorithm of orbital placement because it is an important parameter in the coverage area equation.

- Step 2: Determine the weightage for each involved parameter.
  The satellite should autonomously adjust values of inclination angle, elevation angle, and swath width. The number of satellites in each orbit is determined to reliably cover the whole ground. Weights for each parameter are calculated based on their relative impact on communication and coverage performance.
- Step 3: Use optimization function.
  We suggest using heuristic or metaheuristic algorithms such as Genetic Algorithms (GAs), Particle Swarm Optimization (PSO), Simulated Annealing (SA), and Ant Colony Optimization (ACO) for real-time optimization of satellite coverage. Heuristic and metaheuristic algorithms have several advantages compared to exact or gradient-based methods in satellite coverage optimization because they show outstanding performance in managing complex, high-dimensional, and nonlinear search spaces without requiring derivative information [50–58]. Also, they show a critical advantage for real-time satellite operations because they are computationally efficient and scalable, delivering near-optimal solutions much faster than exact methods. Their flexibility allows them to easily incorporate additional constraints or objectives such as geological models, communication quality, or collision avoidance, while balancing global exploration with local exploitation.
- Step 4: Communication evaluation.
  The link quality and reliable internet connectivity should be evaluated for each coverage optimization selection.
- Step 5: Select optimal coverage area parameters.
  After evaluation of each selection, the most optimal value of inclination angle, elevation angle, and swath width are determined based on the latitude value that is estimated in Equations (3) and (5).

4.3. Expected Results

The proposed solutions are expected to yield significant improvements in both SOP and coverage optimization compared to traditional static or rule-based methods. The results for the two suggested algorithms are expected to improve system resilience, communication performance, and space sustainability. Hence, the proposed frameworks can be used in next-generation LEO networks. The following is a detailed description of the expected results for both algorithms.

- Orbital Placement Based on DRL Solution
  - Improve collision avoidance efficiency by dynamically selecting altitudes and inclinations with lower debris density and risk.
  - Achieve a higher long-term SINR and lower end-to-end delay through learning-based altitude and position adjustment.
  - Maximize a multi-objective reward that balances signal quality, latency, collision risk, and debris avoidance.
  - Enable satellites to adapt in real time to orbital decay, space weather effects, and traffic density, which will reduce the need for manual control.
- Coverage Optimization Based on Metaheuristic Methods
  - Increase total coverage area per satellite or constellation through dynamic adjustment of inclination and swath width.
  - Enhance link reliability and throughput, especially in high-demand areas, by optimizing elevation and footprint overlap.
  - Provide scalable and near-optimal coverage planning that can adapt to different orbital altitudes or regional requirements.

- Enable autonomous satellite configuration, reducing ground control workload and improving operational flexibility.

5. Limitations and Future Works

Although this survey paper presents an extensive survey of SOP and coverage optimization for LEO satellite networks, there were some unavoidable limitations. The space debris cannot be avoided, but it can be mitigated to prevent accidents with the active satellites. The very fast untracked debris objects can cause many satellites to crash due to accidents with those objects, which can lead to stopping internet connectivity in some coverage areas. It is commonly agreed that the lower orbital has a strong signal decay due to atmospheric drag, and the higher orbital has a long signal latency. Furthermore, while the proposed DRL and metaheuristic algorithms offer strong theoretical- and simulation-based performance, their real-world deployment in LEO satellite networks presents several challenges:

- Computational Cost and Energy Consumption: DRL training requires large-scale computation, typically performed on high-performance GPUs or distributed cloud servers. Although the optimization phase occurs at the ground station, frequent retraining for changing orbital dynamics could require significant energy and cost resources.
- Communication Delays and Bandwidth Constraints: Real-time orbital adjustments depend on timely uplink/downlink between satellites and ground stations. The latency of transmitting control commands, especially during high-traffic periods can degrade the responsiveness of AI-driven placement decisions.
- Security and Cyber-Resilience Risks: The use of AI-driven autonomous decision-making increases vulnerability to cyberattacks, data spoofing, or adversarial manipulation of state inputs. Such attacks could lead to unsafe orbital maneuvers or degraded coverage performance if not addressed with robust authentication and encryption mechanisms.
- Explainability and Regulatory Compliance: Regulatory bodies often require transparent decision-making processes for orbital maneuvers. Black-box models like DRL pose challenges for interpretability, which may slow down approval processes or raise trust issues among operators and policymakers.
- Scope of Evaluation: This study primarily focuses on orbital placement and coverage optimization. It does not yet fully integrate power management, inter-satellite routing, or payload optimization, which are critical for comprehensive LEO network performance.

Future Works will focus on developing lightweight AI architectures to reduce computational and energy demands, designing low-latency communication protocols for efficient satellite–ground communication, and integrating real operational telemetry to enhance model adaptability and accuracy under diverse space conditions. Additional efforts will include incorporating cybersecurity-aware AI frameworks to safeguard decision-making against adversarial threats and to explore explainable AI techniques to improve transparency and regulatory compliance. Moreover, the optimization framework will be extended to cover power allocation, inter-satellite routing, and payload scheduling for a more holistic LEO network performance.

6. Conclusions

This paper presents a comprehensive review of the tools and methods used, problems tackled, and results obtained in the previous literature studies in three research areas, which are SOP, coverage optimization, and signal impairment for LEO satellite networks. It introduces the challenges of current internet connectivity and space debris objects. Also,

it proposed solutions for both SOP based on the DRL algorithm and coverage optimization based on the metaheuristic algorithm. The suggested solution utilized multi-input variables such as current altitude, SINR, latency, debris encounter rates, and estimated collision probability. The expected results of the proposed two algorithms will improve system resilience, communication performance, and space sustainability. Hence, the proposed frameworks can be used in next-generation LEO networks. Future work will extend this framework to incorporate adaptive debris-avoidance strategies and enable satellites to adjust their orbits in real-time based on collision risk and network performance metrics.

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